HI ECOMMERCE – Business Insights and Recommendations Report

Product Return Management System | Data Analysis Findings

1. Project Objective

The objective of this analysis is to understand product returns in an e-commerce setting, identify common return patterns, and recommend practical improvements that could reduce avoidable returns and improve the customer experience.

2. Data Analysis Summary

The supplied dataset contains 5,000 orders. There are 2,500 returned orders and 2,500 non-returned orders, giving an overall return rate of 50.00%.

Product category	Returned orders / total	Return rate
Clothing	919 / 920	99.89%
Electronics	614 / 918	66.88%
Home	338 / 847	39.91%
Sports	215 / 586	36.69%
Toys	157 / 536	29.29%
Beauty	163 / 609	26.77%
Books	94 / 584	16.10%

The most frequently recorded return reasons across returned orders were: Wrong item (310; 12.40%), Size/fit issue (298; 11.92%), Changed mind (279; 11.16%), Missing parts (276; 11.04%), Duplicate order (276; 11.04%), Quality issue (275; 11.00%), Damaged (269; 10.76%), Not as described (261; 10.44%), and Late delivery (256; 10.24%).

3. Key Business Insights

- Clothing has a very high recorded return rate (99.89%). This category should be investigated first. Check whether product sizing, fit guidance, product images, descriptions, or data quality are contributing factors.
- Electronics has the second-highest recorded return rate (66.88%). Damaged items are the most frequent recorded reason within this category (84 returns), so packaging, handling, and pre-dispatch checks should be reviewed.
- Wrong item is the most common return reason overall (310 returns). This may indicate opportunities to improve warehouse picking, packing, and order verification.
- Size/fit issues account for 298 returns overall. Clear size charts, fit guidance, and consistent product measurements may help customers choose correctly.

4. Business Recommendations

Problem area	Recommended action

Wrong item	Introduce barcode scanning or a two-step order-item verification before dispatch.
Damaged products	Review protective packaging standards and add a pre-dispatch condition check, especially for electronics.
Size/fit issues	Provide standardised size charts, model measurements, and clearer fit descriptions.
High clothing returns	Review the highest-return clothing SKUs and compare product descriptions, images, sizing information, and customer feedback.
Electronics returns	Add product testing and packaging checks before shipment; investigate damage and defect trends by seller and product.
Data quality and monitoring	Track return rate by product, seller, reason, and time period, and validate unusually high rates before making operational decisions.

5. Expected Benefits

- Fewer avoidable returns and lower reverse-logistics costs.
- Improved order-picking and dispatch accuracy.
- Better customer confidence when selecting products.
- Improved visibility of product, seller, and fulfilment issues.
- More informed decisions using measurable return-rate trends.

6. Important Assumptions and Validation

These findings describe the supplied dataset and should not be treated as verified industry-wide Amazon or India e-commerce statistics. The 99.89% Clothing return rate is unusually high and should be validated at product/SKU level and checked for data or sampling issues before operational decisions are made. Recommendations are hypotheses to investigate, not proven causes.